

Affine-DDPNM: Methodology

Version 5 — revised 2026-08-07

This revision fixes the traction/load sign convention, the orientation of tangential modes, the assumptions for local invertibility and global SPD, and the interpretation of the Galerkin enrichment statement.

We extend the domain-decomposition pore-network method (DD-PNM) of [1] by enriching the interface traction space from a single constant-normal mode to a nine-mode affine representation.

1 Model problem and weak-form convention

Steady incompressible Stokes flow in $\Omega \subset \mathbb{R}^3$:

$$-\nu \Delta \mathbf{u} + \nabla p = \mathbf{0}, \quad \text{in } \Omega, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad \text{in } \Omega, \quad (2)$$

$$\mathbf{u} = \mathbf{0}, \quad \text{on } \Gamma_{\text{wall}}, \quad (3)$$

$$\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = -p_b \mathbf{n}, \quad \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \quad (4)$$

where $\boldsymbol{\sigma}(\mathbf{u}, p) := -p\mathbf{I} + \nu \nabla \mathbf{u}$ is the pseudostress. The inlet/outlet condition prescribes the normal pseudotraction, not the pressure directly: $\mathbf{n} \cdot \boldsymbol{\sigma} \mathbf{n} = -p_b$ does not imply $p = p_b$ pointwise because the normal viscous term contributes. The incompressible weak formulation uses $b_i(\mathbf{v}, q) = -\int_{\Omega_i} q \nabla \cdot \mathbf{v} \, dx$, giving the local discrete system

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U_i \\ P_i \end{bmatrix} = \begin{bmatrix} f_i \\ \mathbf{0} \end{bmatrix}. \quad (5)$$

No pressure stabilisation is used.

2 Domain partition and oriented interface frames

Ω is partitioned into N non-overlapping subdomains $\{\Omega_i\}_{i=1}^N$. Each internal interface γ_k is planar and shared by pores (i_k, j_k) . Choose once a global orthonormal frame $\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$ with $\mathbf{n}_k$ directed from i_k to j_k . Let $\sigma_{i,k} = +1$ for $i = i_k$ and -1 for $i = j_k$.

To match the code's assembled normal direction $-\mathbf{n}_{i,k}$ and tangential directions $\sigma_{i,k} \mathbf{t}_{k,\nu}$, define **oriented inward-traction directions**

$$\mathbf{d}_{i,k,n} = \mathbf{n}_{i,k}, \quad \mathbf{d}_{i,k,t_\nu} = -\sigma_{i,k} \mathbf{t}_{k,\nu} \quad (\nu = 1, 2). \quad (6)$$

The minus sign in the tangential definition is an orientation choice; changing it together with the corresponding coefficient changes no traction space. Crucially,

$$\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta} \quad (\beta = n, t_1, t_2). \quad (7)$$

Half-span-normalised in-plane coordinates $s_k(\mathbf{x}) = ((\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,1})/a_k$, $t_k(\mathbf{x}) = ((\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,2})/b_k$, with scalar shapes $\varphi_0 = 1$, $\varphi_s = s_k$, $\varphi_t = t_k$, $\mathcal{M} = \{0, s, t\} \times \{n, t_1, t_2\}$.

3 Affine traction Ansatz and code-compatible load sign

Assumption 1 (Local algebraic stability). *For every pore i :*

- (a) A_i is SPD (guaranteed e.g. by a positive-measure no-slip wall segment);
- (b) B_i has full row rank (discrete inf-sup);
- (c) the constant function 1 belongs to the local pressure space.

The unknown coefficients parameterize the **inward** traction:

$$-\boldsymbol{\sigma}(\mathbf{u}_i, p_i) \mathbf{n}_{i,k} = \sum_{(\alpha, \beta) \in \mathcal{M}} \lambda_{k, \alpha, \beta} \varphi_\alpha \mathbf{d}_{i,k, \beta}. \quad (8)$$

The physical outward traction in the weak form is therefore $\boldsymbol{\sigma} \mathbf{n}_{i,k} = -\sum \lambda \varphi \mathbf{d}_{i,k, \beta}$.

The primitive load vector for mode (α, β) is

$$[\mathbf{f}_{i,k}^{(\alpha, \beta)}]_j = - \int_{\gamma_k} \varphi_\alpha \mathbf{d}_{i,k, \beta} \cdot \boldsymbol{\phi}_j^{(i)} dS. \quad (9)$$

The leading minus sign is essential: it converts the inward-traction Ansatz into the physical outward-traction boundary integral. For the normal mode this gives the assembled direction $-\mathbf{n}_{i,k}$; for the tangential modes (6) gives $+\sigma_{i,k} \mathbf{t}_{k,\nu}$, exactly the convention used in `affine_face_basis.py`.

At an inlet/outlet port only the constant normal mode is retained with prescribed coefficient $\lambda_{i,\gamma}^k = p_b$.

4 Local response and compliance map

For each primitive mode solve

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U_{i,k}^{(\alpha, \beta)} \\ X_{i,k}^{(\alpha, \beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,k}^{(\alpha, \beta)} \\ \mathbf{0} \end{bmatrix}. \quad (10)$$

Let $n_{u,i}$ denote the number of local velocity DOFs in Ω_i . Let $L_i \in \mathbb{R}^{n_{u,i} \times n_i}$ collect all $n_i = 9m_i^u + m_i^k$ primitive load columns. Define

$$M_i = B_i A_i^{-1} B_i^\top, \quad H_i = A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}. \quad (11)$$

Then $U_i = H_i L_i \boldsymbol{\lambda}_i$ and the local compliance matrix is

$$\mathbf{G}_i = L_i^\top H_i L_i \in \mathbb{R}^{n_i \times n_i}, \quad (12)$$

symmetric positive semidefinite, with $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = U_i^\top A_i U_i \geq 0$.

For mode (α, β) on interface γ_k , the **weighted velocity moment** is

$$M_{i,k}^{(\alpha, \beta)}(U_i) = \int_{\gamma_k} \varphi_\alpha U_{i,h} \cdot \mathbf{d}_{i,k, \beta} dS. \quad (13)$$

Only $M_{i,k}^{(0,n)}$ is the volumetric flow rate; the other eight are weighted velocity moments (linear normal flux moments and tangential velocity moments). From the load convention, $[\mathbf{G}_i \boldsymbol{\lambda}_i]_{k, \alpha, \beta} = -M_{i,k}^{(\alpha, \beta)}(U_i)$. $\mathbf{G}_i$ is partitioned into $\mathbf{G}_i^{uu}$ (internal–internal) and $\mathbf{G}_i^{uk}$ (internal–boundary).

5 Global Schur complement assembly

Continuity of all nine weighted velocity moments, $M_{i,k}^{(\alpha,\beta)} + M_{j,k}^{(\alpha,\beta)} = 0$, together with Boolean assembly gives

$$\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}, \quad \mathbf{S} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k. \quad (14)$$

$\mathbf{S} \in \mathbb{R}^{9m \times 9m}$ is symmetric by construction.

Remark 1 (Conditions for SPD). $\mathbf{S}$ is positive definite under the following conditional hypotheses (see the companion mathematical properties document for full statements and proofs):

1. **Local stability:** Assumption 1.
2. **Local kernel:** $\text{rank}(H_i L_i) = n_i - 1$ for every pore. This condition cannot be reduced to per-face linear independence of the nine modes; it additionally excludes cross-interface dependencies and dependencies introduced by the incompressible projection H_i .
3. **Graph anchoring:** every connected component of the pore adjacency graph contains at least one inlet/outlet boundary port.

Floating basins (pores without a solid-wall facet) violate Assumption 1(a) and are merged into neighbours before assembly. This restores local coercivity but does not supply condition (3); the two issues are logically distinct.

6 Algorithm

Algorithm 1: Affine-DDPNM solver

Input: Subdomain meshes $\{\Omega_i\}$, FE pairs $(\mathbf{V}_{i,h}, Q_{i,h})$, prescribed boundary coefficients $\boldsymbol{\lambda}^k$.

Output: Unknown interface coefficients $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$, local fields $\{(U_i, P_i)\}$, weighted velocity moments, and residuals.

Offline library — for each pore $i = 1, \dots, N$ (in parallel):

1. Assemble A_i, B_i .
2. For each internal interface $\gamma \in \mathcal{U}_i$ and each mode $(\alpha, \beta) \in \mathcal{M}$: assemble $\mathbf{f}_{i,\gamma}^{(\alpha,\beta)}$ via (9), solve (10) for $(U_{i,\gamma}^{(\alpha,\beta)}, X_{i,\gamma}^{(\alpha,\beta)})$.
3. For each boundary port $\gamma \in \mathcal{K}_i$: assemble and solve the single $(0, n)$ mode.
4. Form L_i , compute $\mathbf{G}_i = L_i^\top H_i L_i$ via (12). As a diagnostic, inspect the singular values of $H_i L_i$ or the eigenvalues of $\mathbf{G}_i$: confirm one numerically null eigenvalue (within a stated tolerance) with eigenvector aligned with $\mathbf{1}_{i,0n}$ (the local-kernel condition in Remark 1, item 2).
5. Accumulate global Schur system: $\mathbf{S} += (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u$, $\mathbf{F} -= (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k$.

Global solve: Check symmetry and positive definiteness of $\mathbf{S}$ (Cholesky factorization or smallest eigenvalue). Solve $\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}$ by sparse direct solver.

Reconstruction — for each pore $i = 1, \dots, N$ (in parallel):

$$U_i = \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha, \beta} \lambda_{i,\gamma,\alpha,\beta} U_{i,\gamma}^{(\alpha,\beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i,\gamma}^k U_{i,\gamma}^{(0,n)},$$

and analogously for P_i with the pressure responses $X_{i,\gamma}^{(\cdot)}$.

Verification: Check local mass residuals, interface moment residuals $M_{i,\gamma}^{(\alpha,\beta)} + M_{j,\gamma}^{(\alpha,\beta)}$, and global boundary flux balance independently.

7 Relation to classic DD-PNM and scope of enrichment

Let $E_{P0} : \mathbb{R}^m \rightarrow \mathbb{R}^{9m}$ embed each classic coefficient into the constant-normal slot. The classic DD-PNM system is recovered by restricting **both trial and test spaces**:

$$\mathbf{S}_{P0} = E_{P0}^\top \mathbf{S} E_{P0}, \quad \mathbf{F}_{P0} = E_{P0}^\top \mathbf{F}.$$

The full affine solution $\boldsymbol{\lambda} = \mathbf{S}^{-1} \mathbf{F}$ generally does *not* have the classic solution as its constant-normal subvector, because the eight additional test-space constraints alter the Galerkin projection.

The valid enrichment statement is conditional and reference-dependent. If a well-defined full interface-traction Schur system $\hat{\mathbf{S}} \boldsymbol{\xi} = \hat{\mathbf{F}}$ is introduced with the same local pressure spaces and sign conventions (the broken-pressure C-DD system of the companion mathematical properties document), then the affine system is its Galerkin restriction and the affine error is no larger than the classic error in the $\hat{\mathbf{S}}$ energy **seminorm**. This does *not* imply monotonic improvement of velocity L^2 , pressure L^2 , broken H^1 , or outlet flux errors.

No theorem in the present analysis establishes equality between that broken-pressure traction system and the monolithic continuous-pressure Taylor–Hood system. The “Exact FE-trace Schur” computation is a primal Schur reduction of the monolithic system and therefore verifies its own algebraic elimination, not the missing equivalence.

8 Computational cost

A pore with m_i^u internal ports and m_i^k boundary ports requires $n_i = 9m_i^u + m_i^k$ primitive solves. Offline cost grows approximately $9\times$ relative to the one-mode-per-interface classic method. The global unknown count is $9m$; sparse factorization cost depends on graph fill and cannot be inferred from the unknown-count factor alone. For multi-boundary-condition studies the offline library is built once.

References

- [1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.